

UNCLASSIFIED



U.S. Coast Guard Coast Guard Cyber Command

Taking on the Dark Fleet
...in Cyberspace!



Kenny Miltenberger
Commander, U.S. Coast Guard
Commanding Officer, 2003 Cyber Protection Team

Shane Cancilla
Network Engineer, 2003 Cyber Protection Team

UNCLASSIFIED



whoami's



Kenny



Shane



UNCLASSIFIED

We do cyber



UNCLASSIFIED

Source: [The Claw of Knowledge](#)



UNCLASSIFIED

U.S. Coast Guard Cyber Protection Teams

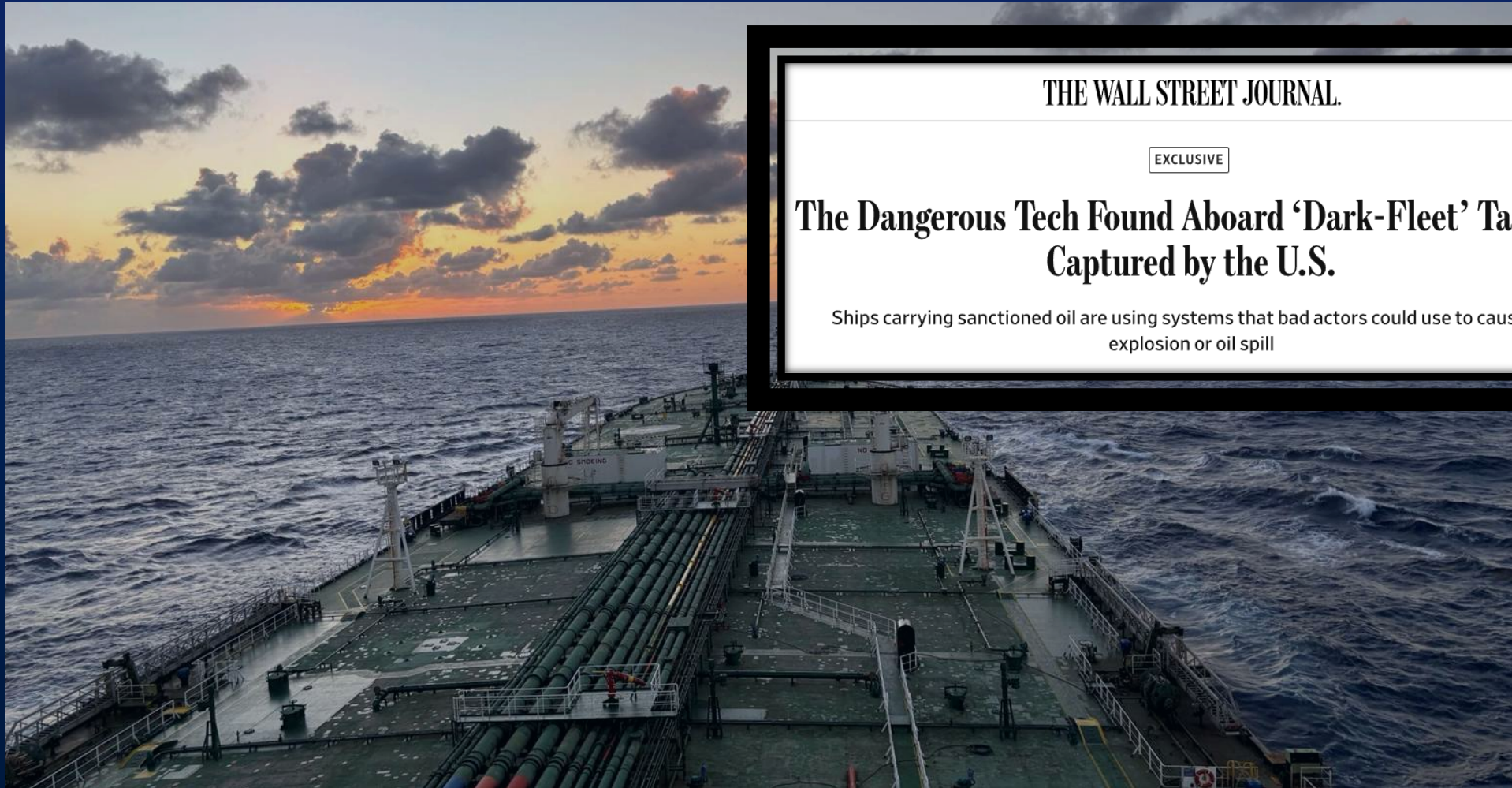


UNCLASSIFIED

Last year



This year



THE WALL STREET JOURNAL

EXCLUSIVE

The Dangerous Tech Found Aboard 'Dark-Fleet' Tankers Captured by the U.S.

Ships carrying sanctioned oil are using systems that bad actors could use to cause an explosion or oil spill



The Dark Fleet

- AKA: shadow, ghost, or dark fleet
- Disaggregated
- Avoid sanctions & regulations
- False flags and open registries
- In 2024 US Congressional Research Service estimates 1600 ships carrying sanctioned oil



[More reading: The Global Oil Tanker Market: An Overview as It Relates to Sanctions | Congress.gov | Library of Congress](#)



Dark Fleet Crackdown

- December 2025 led by the US
- Shortly followed by interdictions from: India, Belgium, France, Sweden, UK, Finland, Estonia, Germany, and others.
- Over 19 physical interdictions globally since December 2025.
(up from <5 in the previous 10 years total)



Example of an interdicted vessel, M/T Sophia, was operating in international waters and conducting illicit activities in the Caribbean Sea. [Source](#)



Safety of Dark Fleet Vessels

- Dark Fleet Vessels are not compliant with rules and regulations or general due diligence to secure IT and OT networks aboard.
- Tanker lifespans.
- Proliferation of high speed, low cost satellite communications.
- Example: inert gas generators.



[Source: Inert Gas System Requirements on Ships 2025 Guide](#)



UNCLASSIFIED

?



UNCLASSIFIED

Solution: Integrate Cyber

- Cyber Control Teams
- Cyber Positive Control (POSCON)
- Guiding principals: protect our boarding team, protect the vessel, and protect the vessel's crew.
- Being helpful nerds to our boarding teams.



Getting out of your comfort zone



Jake getting hoisted

Operational

PPE

Camping gear

MREs

Technical

Enterprise Networks
vs. Small-Network Oil
Tankers



Alan climbing Jacob's Ladder



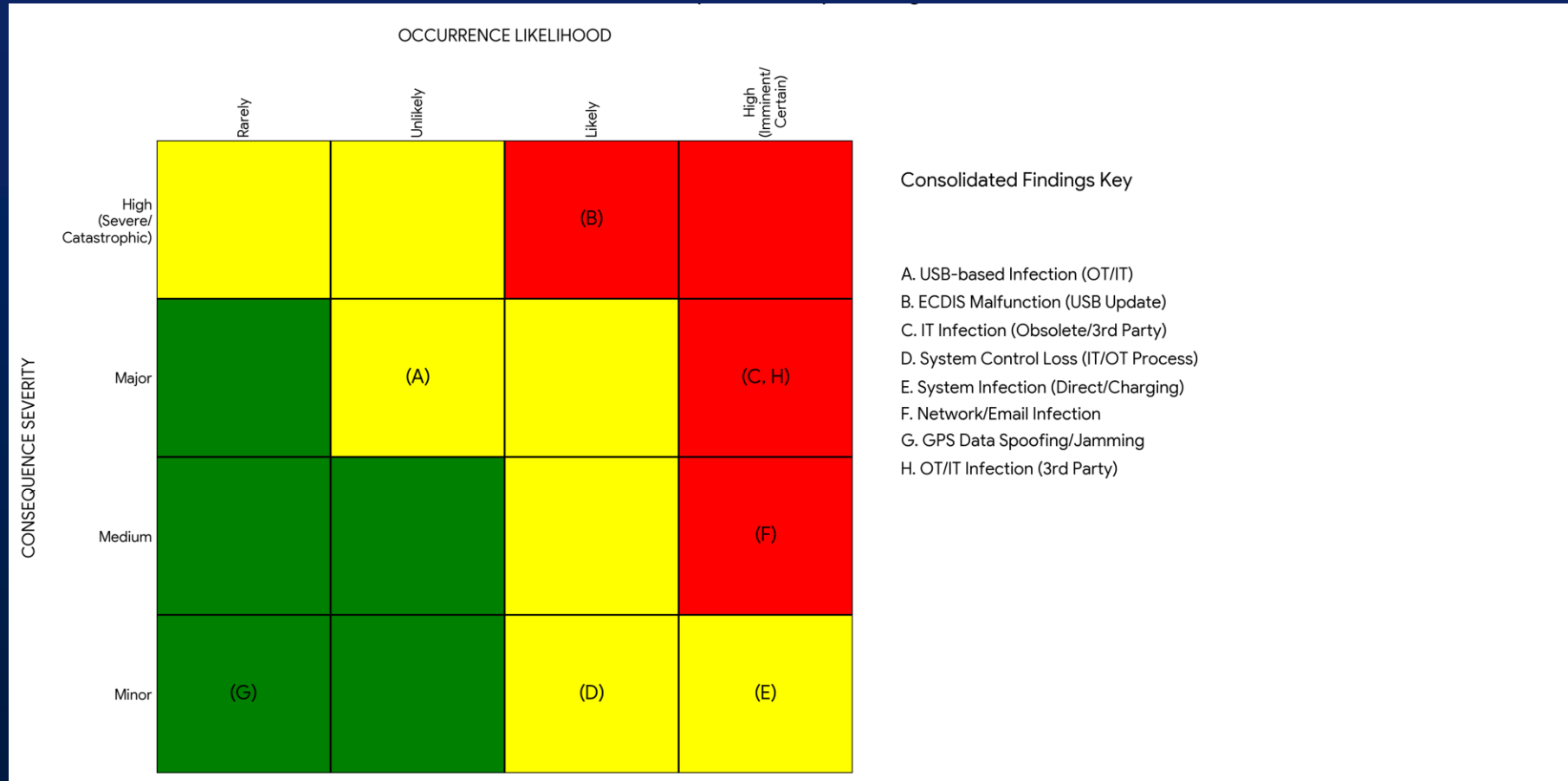
Mike eats an MRE onboard a seized tanker



Findings



Cybersecurity of Dark Fleet Vessels



Consolidated Findings of Dark Fleet Vessel Cybersecurity Assessment



Dark Fleet Networking

- **Satellite Internet:**

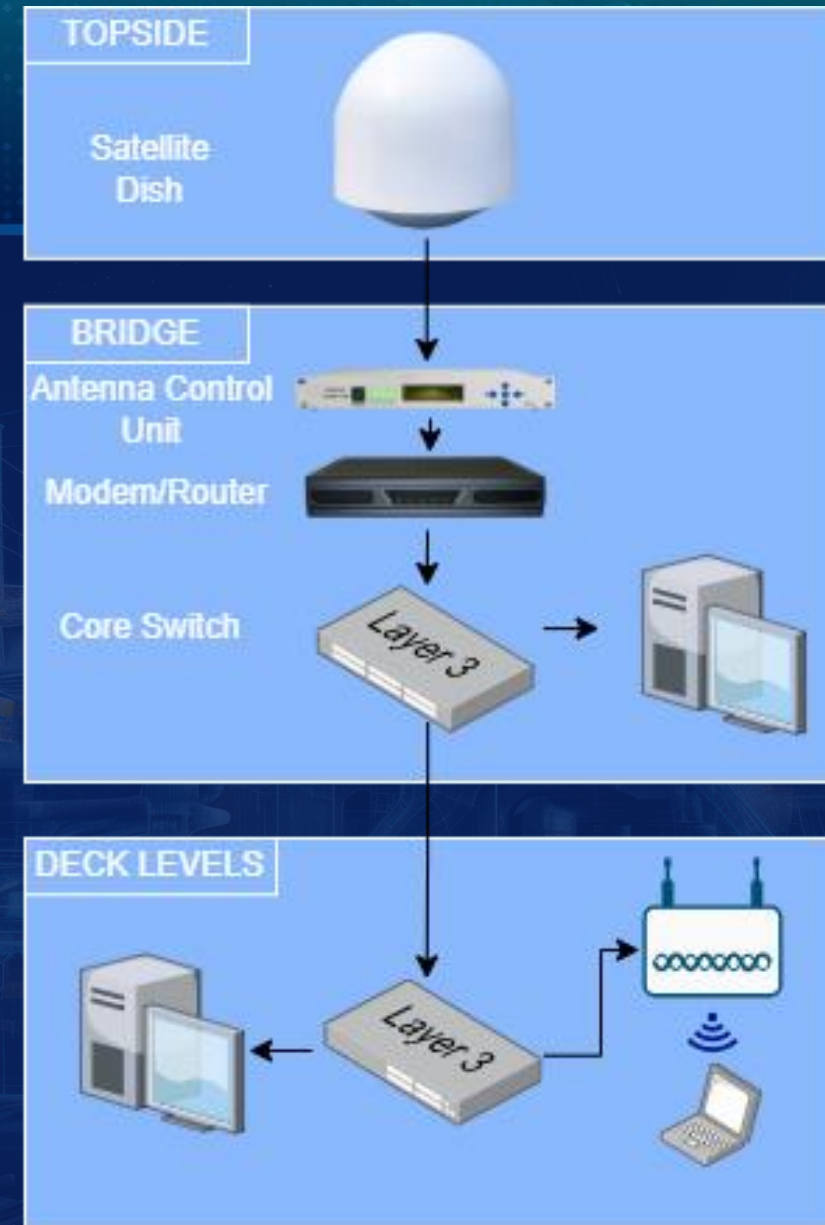
- KU Band
- StarLink

- **Onboard Networking:**

- Modem
- Core Switch
- User Switches/WAPs



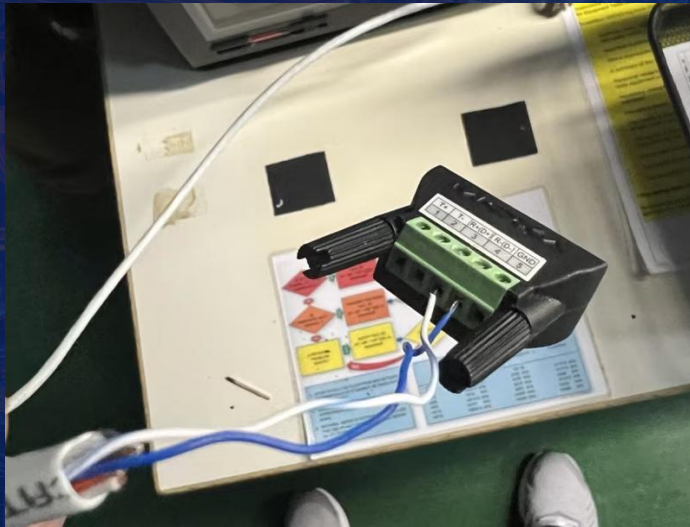
[Source: Maritime IT Operations with Satellite Communication and Starlink](#)



IT/OT Segmentation

OT / IT Integration: Bad cyber-hygiene, and OT/IT integration make this more and more difficult.

Variable Bad cyber-hygiene, and OT/IT integration make this more and more difficult.



Ethernet to Serial on AIS



Engineering Control Room CAN to Ethernet Gateways

Malware & Anti-Virus

- **Malware:** Compromised segmentation through foreign USB devices is a well known and documented issue onboard Dark Fleet Vessels...
- **Exploitable propagation methods observed:**
 - SMB
 - RDP
 - Weak Credentials (like rockyou.txt weak), shared, or not used at all
 - Disabled anti-virus (or disabling after running a malicious executable)
 - Outdated OSs
 - Shared removable media devices

Event	Count
Confirmed Malware Detections	17
Threat history cleared	344
Remediation Failures	2
Protection Disabled	5
Configuration Changed	2,268

Fleet-Wide Malware / AV Statistics

Malware & Anti-Virus

- **Malware:** Multiple observed cases of malicious software living on host workstations and public vessel servers
 - Foreign USB introduction to air gapped networks is a known issue.
- **MusaLLaT.exe:** USB Propagating malware execution attempted, could endanger air gapped networks.
 - Changes local DNS records to prevent EDR solutions from accessing public servers.
 - Disables UAC, TaskManager, Cmd Prompt, Windows System Restore, and Regedit.
 - Firewall modifications to allow inbound TCP/139.
 - Attempts to write legacy autorun.inf file to D:\, and hides it.
- **Trojans:** Numerous trojans, some targeting PRC social practices

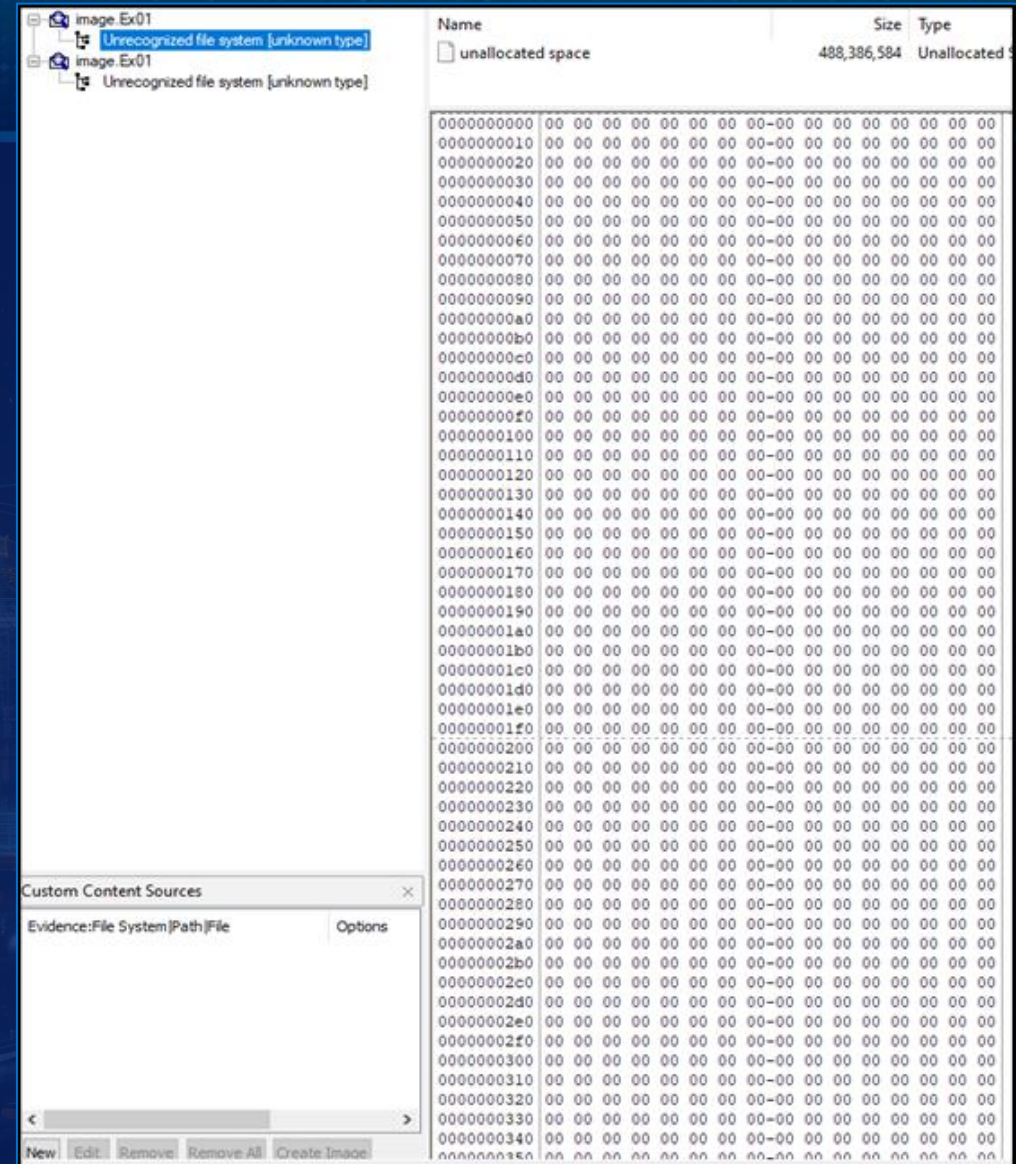
Source: MusaLLaT.exe, 00000001.00000003.1172675567.00000000007A1000.00000004.00000020.00020000.00000000.sdmp	Binary or memory string: \??\D:\autorun.inf
Source: MusaLLaT.exe, 00000001.00000003.1173471114.00000000007A1000.00000004.00000020.00020000.00000000.sdmp	Binary or memory string: \??\D:\autorun.inf

Process:	C:\Users\user\AppData\Roaming\MusaLLaT.exe
File Type:	ASCII text, with CRLF line terminators
Category:	dropped
Size (bytes):	4722
Entropy (8bit):	4.451855525052748
Encrypted:	false
SSDEEP:	96:uRrDPgyaQckg9Kqkhs5xiPeNrkmtkv/hnCHj2:uRrBaQcxzmjPeNrO4a
MD5:	D62DBEA82A3B61B280E9AF18FF7A3E2B
SHA1:	FABEA61665D61E9D099E463E5A5F9FCB069AF2BF
SHA-256:	DBAF01F64A24A2080A7ED8C37A50EB9B312C0C6410CFA636862DA5D9C682C468
SHA-512:	FC162182E1E560906E011385A2DEDE3A043A4D80B3CC50BC149DD7DED8DFE08BF83E2C178E1A73DBC9263BCC76F6887716A27A4598D5577DFA24554B3A25F363
Malicious:	true
Reputation:	low
Preview:	127.0.0.1 threatsense.net..127.0.0.1 www.threatsense.net..127.0.0.1 www.zma.com.ar..127.0.0.1 zma.com.ar..127.0.0.1 store.ca.com..127.0.0.1 avira.com..127.0.0.1 www.antivir.com..127.0.0.1 antivir.com..127.0.0.1 www.antivir.com.tr..127.0.0.1 www.avg.com..127.0.0.1 avg.com..127.0.0.1 www.scanwith.com..127.0.0.1 scanwith.com..127.0.0.1 www.avast.gen.tr..127.0.0.1 avast.gen.tr..127.0.0.1 www.avast.com..127.0.0.1 vast.com..127.0.0.1 forum.avast.com..127.0.0.1 www.nod32.com..127.0.0.1 nod32.com..127.0.0.1 novirusthanks.org..127.0.0.1 novirusthanks.org..127.0.0.1 vscan.novirusthanks.org..127.0.0.1 virustotal-upload.er.en.softonic.com..127.0.0.1 virscan.org..127.0.0.1 pandasecurity.com..127.0.0.1 www.arcabit.com..127.0.0.1 arcabit.com..127.0.0.1 www.arcabit.pl..127.0.0.1 arcabit.pl..127.0.0.1 www.freedweb.com..127.0.0.1 freedweb.com..127.0.0.1 www.drweb.com..127.0.0.1 drweb.com..127.0.0.1 www.drweb-online.com..127.0.0.1 drweb-online.com..127.0.0.1 www.es

MusaLLaT Dynamic Analysis Artifacts

Technical Findings – Remote Disk Wiping

- **Remote Access:**
 - TeamViewer
 - RemoteDesktop
 - ScreenConnect
- **System Tampering:** Remote access to workstation onboard then PowerShell from there to zero-fill server hard drives.



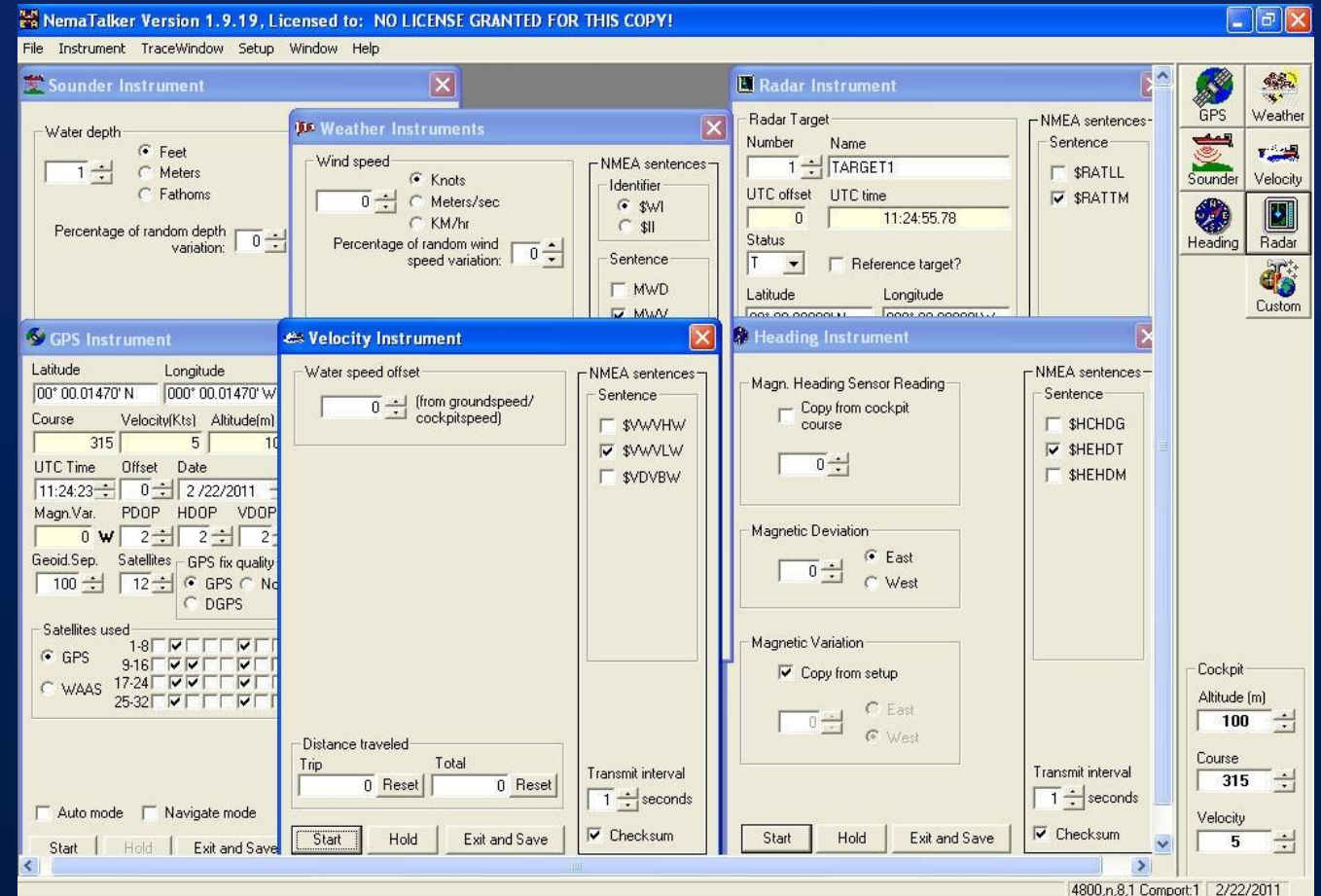
Navigation Systems Tampering – AIS manipulation

- **Antenna Tampering:** Apparent attenuation cages around pilothouse antennas, suspected to support AIS spoofing activities by cutting off receipt of legitimate GPS signals.



Vessel Position

Simulation: Use of laptops with specialized software to spoof NMEA data outputs through serial connection to AIS.



Example NMEA manipulation software

Source: [NemaTalker 1.9 Download \(Free trial\) - NemaTalker.exe](#)



Multiple AIS units & NMEA 0183

Tampering: Spliced serial connections between GPS & bridge navigation systems.

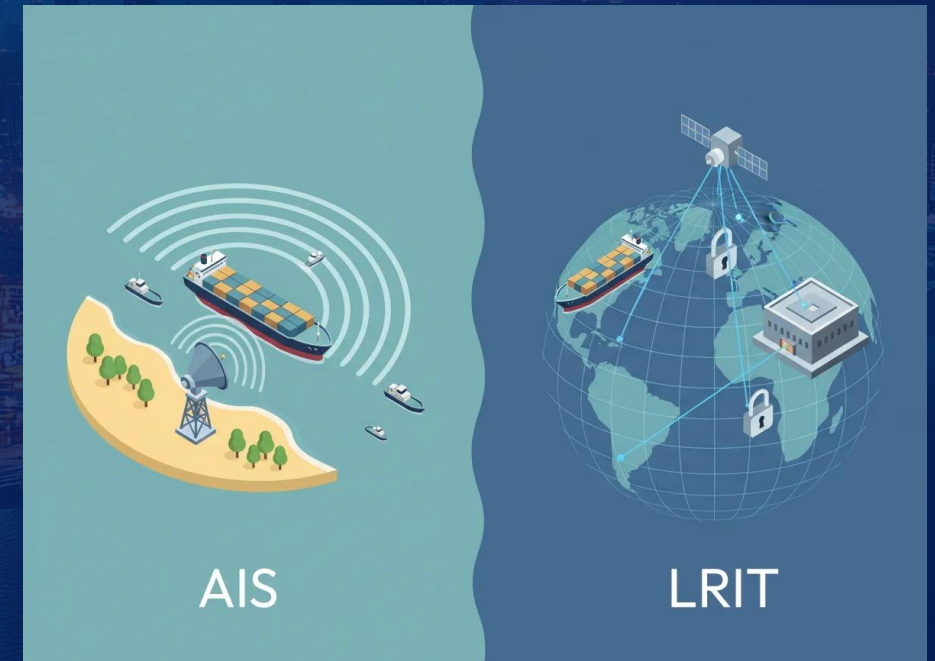


Physical selector between two AIS units of different brands



Navigation Systems Tampering – Long Range Identification & Tracking

- **LRIT:** Vessels over 300 gross tons are mandated by the International Maritime Organization (IMO) to transmit their identity & location every 6 hrs to one of 4 bodies:
 - Coastal State
 - Flag State
 - Port State
 - SAR Authorities
- Typically built on INMARSAT-C
 - Closed, encrypted, private communications for coastal and flag state monitoring.
- Spoofing from a legal & safety prospective is much more serious, but typically harder in practice.
- To spoof, simulate external GPS or tamper with firmware



AIS vs LRIT

Navigation Systems Tampering – Long Range Identification & Tracking

- **Firmware:** we observed shared firmware for INSMARSAT-C terminals
- **GPS Spoofing:** observed indicators of when the vessel began AIS spoofing, it spoofed their LRIT position as well, indicating there's likely no dedicated GPS unit unlike most newer models.
- **Impact:** Evades flag and coastal state level tracking which further aid in evading sanctions & legal prosecution. Could also negatively impact SAR cases by not being properly identified as a potential "good Samaritan."



Generic LRIT Terminal

Collateral Findings

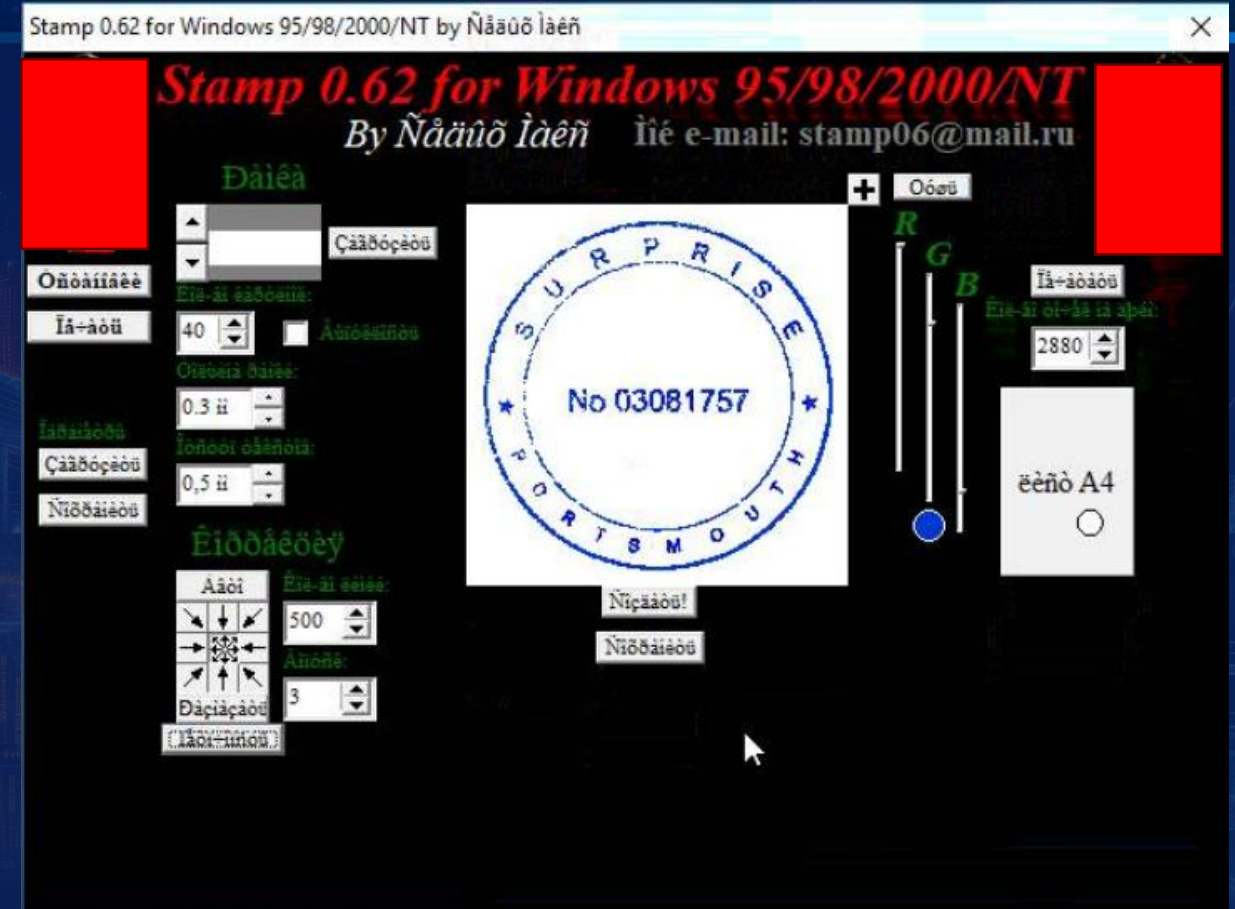


Strange Programs – Stamp 0.62

Stamp 0.62: Russian Windows Executable for custom stamp generation to support rapid changing of vessel names and IMO numbers for use on official documentation



Stamp 0.62



Re-Created Example of Vessel Stamp

Acknowledgements



LTJG Noah Soto –
holistic dark fleet
analysis lead.

